

Austin Ellis

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EDUCATION

- **California State University, Northridge** 2023 – 2026
Master of Science in Chemistry Los Angeles, USA
- **University of California, San Diego** 2018 – 2022
Bachelor of Science in Pharmacological Chemistry San Diego, USA

RESEARCH EXPERIENCE

- **Research Student Internship, Oak Ridge National Laboratory** May 2026 – Present
Advisors: Jingsong Huang and Ilia Ivanov
 - Conducted computational research on acetonitrile reactivity under pressure, in alcohol mixtures, and on alkali metals using density functional theory, machine-learned interatomic potentials (MLIPs), and automated reaction-path searches.
 - Developed [ReactionFlow](#), an open-source tool that turns bond changes seen in MLIP molecular dynamics into converged reaction paths, and set up reaction-discovery workflows on Perlmutter and Frontier.

Selected Research Projects

- **Benchmarking Pretrained Interatomic Potentials Across Reaction Pathways** | Manuscript in preparation
 - Benchmarked 15 pretrained MLIPs on 874,176 structures from 9,017 reactions (DORTS-9K), plus Transition1x and compressed acetonitrile.
 - Compared energy, force, and Hessian errors against DFT to assess accuracy and transferability across reaction types.
- **Pressure-Induced Polymerization of Acetonitrile**
 - Ran reactive MLIP molecular dynamics and over 500 NEB calculations on compressed β -acetonitrile, tracing a saddle-validated route from hydrogen abstraction to tetramer growth.
 - Showed that the radical intermediates survive only in spin-polarized DFT, then fine-tuned UMA on spin-resolved PBE-D3 data, cutting held-out force error by 43%.
- **Acetonitrile Cyclotrimerization in Methanol and on Metal Interfaces**
 - Mapped DFT and AutoMeKin reaction routes from acetonitrile-methanol mixtures to the two experimentally observed products, trimethyltriazine and 4-amino-2,6-dimethylpyrimidine.
 - Modeled sodium and copper-mediated trimerization at interfaces, connecting an eight-step hydrogen-relay pathway to the same aminopyrimidine.
- **Graduate Research Assistant, CSUN** | Advisor: Maosheng Miao | [M.S. Thesis](#) January 2023 – May 2026
 - Conducted computational research on high-pressure materials, electrides, superconducting hydrides, and electron localization using density functional theory, crystal-structure prediction, and machine learning.
 - Developed workflows combining VASP, CALYPSO structure searches, hybrid-functional calculations, and electronic-structure analyses including ELF, Bader charge, projected DOS, and ICOHP.

Selected Research Projects

- **Prediction of Electron Localization Functions from Superposed Atomic Densities** | [Paper](#)
 - Designed a periodic 3D residual network using circular-padding convolutions, attention modules, and a peak-aware composite loss to recover sparse interstitial localization features under high pressure.
 - Trained on 326,009 unary and binary structures spanning 89 elements and evaluated on 50,000 metal-only superhydride templates, running $345\times$ faster generation than DFT.
- **Molecular Hydrogen Metallization under High Pressure** | [Paper](#)
 - Calculated PBE and HSE06 electronic structures for candidate molecular hydrogen phases to track pressure-induced band-gap closure.
 - Analyzed interstitial wavefunction localization near the Fermi level and identified emerging electride behavior preceding molecular dissociation.
- **Chemical-Template-Guided Discovery of Complex Metal Superhydrides** | [Paper](#)

- Co-assembled a superhydride training set by expanding structural prototypes across s-d border metals and multiple pressures for training of ensemble machine-learning models to screen candidate metal lattices
- Performed chemical-template-guided CALYPSO searches and stability screening that identified 13 structural prototypes and 31 stable metal superhydrides.
- **High-Pressure Redox Chemistry of Iron in Deep Earth Systems** | [Paper](#)
- Mapped pressure-dependent phase stability across Fe-p-block compositions at 0, 150, and 300 GPa using CALYPSO structure searches and HSE06 convex-hull calculations.
- Quantified the pressure-driven reversal of Fe redox behavior through hybrid-functional electronic-structure and Bader charge analyses.
- **Electronic Structure of Superconducting Molecular Hydride NaKH₁₂** | [Paper](#)
- Performed band-structure, projected density-of-states, and crystal-orbital analyses to determine how NaK interstitial electrons metallize the H₂ framework in NaKH₁₂.
- Tracked pressure-dependent ELF evolution in the metal sublattice to connect interstitial electron localization with H-H bond weakening and superconducting behavior.
- **Tunable Electrides in Vacancy-Engineered Monolayer Transition Metal Dichalcogenides** | [Paper](#)
- Analyzed electride-like interstitial states in monolayer TMDCs using ELF, charge-density differences, and projected electronic band structures.
- Examined chalcogen vacancies as a route to tune quasiautom states and assisted with H-adsorption calculations relevant to the hydrogen evolution reaction.

TECHNICAL EXPERIENCE

- **HPC Cluster Manager, College of Science and Mathematics, CSUN** *August 2023 – May 2026*
 - Administer, monitor, and optimize the high-performance computing environment, ensuring consistent uptime and efficient job scheduling.
 - Improved cluster efficiency 48 × by enabling node oversubscription and optimizing cores per submitted job.
 - Administer two GPU cluster nodes with 4 × A100 GPUs each, including driver and kernel upgrades, remote power-cycle recovery, and proactive troubleshooting to maximize uptime and reliability for large-scale HPC and ML training.
 - Prepare HPC accounts and environments for CSUN faculty and students performing quantum-chemistry calculations for research and coursework.

PUBLICATIONS

- [6] **Prediction of Electron Localization Functions from Superposed Atomic Densities for Accelerated Superhydride Discovery**
A. Ellis, S. Ellis, M. Miao
Journal of Materials Informatics. **2026**, 6, 45. DOI: [10.20517/jmi.2026.44](#)
- [5] **Emerging Electride Behavior and Metallization in Molecular Hydrogen under High Pressure**
A. Ellis, S. Ellis, A. Pandit, M. Miao
The Journal of Physical Chemistry Letters. **2026**, 17(29), 8198–8205. DOI: [10.1021/acs.jpcclett.6c01415](#)
- [4] **Pressure-induced Redox Reversal of Iron and the Distribution of Elements in Deep Earth**
X. Wang, X. Feng, J. Li, Y. Lv, A. Ellis, S. Scott, A. Pandit, D. Khodagholian, R. J. Hemley, M. G. Jackson, F. J. Spera, S. A. T. Redfern, M. Miao
Proceedings of the National Academy of Sciences. **2025**, 122, e2414911122. DOI: [10.1073/pnas.2414911122](#)
- [3] **Batch Discovery of Complex Metal Superhydrides via an Effective Machine Learning Method Structured by Chemical Template Concept**
Y. Sun, A. Ellis, X. Chen, M. Miao
Journal of the American Chemical Society. **2025**, 147(44), 40407–40419. DOI: [10.1021/jacs.5c11731](#)
- [2] **Unlocking the Origin of High-Temperature Superconductivity in Molecular Hydrides at Moderate Pressures**
W. Zhao, A. Ellis, D. Duan, H. Wang, Q. Jiang, M. Du, T. Cui, M. Miao
Advanced Functional Materials. **2025**, 35, 2415910. DOI: [10.1002/adfm.202415910](#)
- [1] **Constructing Tunable Electrides on Monolayer Transition Metal Dichalcogenides**
Y. Sun, A. Ellis, S. Diaz, W. Li, and M. Miao
The Journal of Physical Chemistry Letters. **2024**, 15(23), 6174–6182. DOI: [10.1021/acs.jpcclett.4c01263](#)

- [3] **Accuracy and Limits of Pretrained Interatomic Potentials Along Molecular Reaction Pathways**
A. Ellis, R. Morelock, S. Bagchi, A. M. dos Santos, M. Goswami, J. Jakowski, J. Huang, P. Ganesh, I. Ivanov
- [2] **Stimulus-Directed Oligomerization and Polymerization of Acetonitrile**
I. N. Ivanov, A. Ellis, P. Christakopoulos, A. Puresky, C. Rouleau, J. Huang, J. Jakowski, M. Goswami, P. Ganesh, B. Sumpter
- [1] **Pressure-induced Core Reactivity of Cesium in Cs–Ba–O Compounds**
S. Ellis, A. Ellis, L. Wyke, M. Miao

SELECTED PRESENTATIONS

Oral Presentations:

- A. Ellis et al.: *Machine-Learned Discovery of Catalyst-Assisted Reaction Pathways and Kinetics in Acetonitrile*. Invited Talk, NTI Group Meeting, Center for Nanophase Materials Sciences, Oak Ridge National Laboratory, September 2026.
- A. Ellis et al.: *Taming Inert Acetonitrile: Probing Pressure-Induced Reactivity and Transition-State Searches with Machine-Learned Interatomic Potentials*. Talk, Neutron Scattering Division, Oak Ridge National Laboratory, August 2026.
- A. Ellis and M. Miao: *Symmetry Aware Prediction of Electron Localization Functions from Superposed Atomic Densities*. Talk at the 2026 American Physical Society March Meeting (APS), Denver, Colorado, March 2026.
- A. Ellis and M. Miao: *Chemistry of High Pressure Materials*. Invited presentation at the CSUN Research Showcase, Department of Chemistry and Biochemistry, California State University, Northridge, October 2025.
- A. Ellis, Y. Sun, S. Diaz, and M. Miao: *Applications of OPTIMADE in Structure Screening within High-Pressure Regimes*. Talk at the 2023 CECAM Open Databases Integration for Materials Design Workshop, Lausanne, Switzerland, June 2023.

Poster Presentations:

- A. Ellis et al.: *Taming Inert Acetonitrile: Probing Pressure-Induced Reactivity and Transition-State Searches with Machine-Learned Interatomic Potentials*. Poster presentation at the CNMS User Meeting, Center for Nanophase Materials Sciences, Oak Ridge National Laboratory, August 2026.
- A. Ellis et al.: *Decoding the Signatures and Pathways to a Transition State*. Poster presentation for the Nonequilibrium and Emergent Transients in Advanced and Soft Materials (NEAT) Initiative, Laboratory Directed Research and Development Program, Oak Ridge National Laboratory, July 2026.
- A. Ellis and M. Miao: *Symmetry Aware Prediction of Electron Localization Functions from Superposed Atomic Densities*. Poster presentation at the 2026 Gordon Research Conference on Enabling Multifunctionality through AI Informed Material Discovery and System Design, Ventura, California, January 2026.
- A. Ellis, Y. Sun, X. Chen, and M. Miao: *An Effective Machine Learning Frame for Materials Discovery Structured by a Chemical Concept*. Poster presentation at the 2025 NeurIPS AI4Mat Workshop, San Diego, California, December 2025.
- A. Ellis and M. Miao: *Symmetry Aware Prediction of Electron Localization Functions from Superposed Atomic Densities*. Poster presentation at the 2025 NeurIPS AI4Mat Workshop, San Diego, California, December 2025.
- A. Ellis, Y. Sun, S. Diaz, W. Li, and M. Miao: *Constructing Tunable Electrides on Monolayer Transition Metal Dichalcogenides*. Poster presentation at the 2025 American Physical Society Global Summit (APS), Anaheim, California, March 2025.
- A. Ellis, Y. Sun, S. Diaz, and M. Miao: *Core Reactivity of Cs Polyoxides Under Pressure*. Poster presentation at the 2023 UCLA Theoretical Chemistry Conference (TheoChem), Los Angeles, California, October 2023.

Co-authored Presentations:

- M. Miao (presenter), Y. Sun, and A. Ellis: *From the electride concept to an efficient machine learning method*. Talk at the 2026 American Physical Society Global Summit (APS), Denver, Colorado, March 2026.
- S. Scott (presenter), X. Wang, X. Feng, J. Li, Y. Lv, A. Ellis, A. Pandit, D. Khodaghlian, R. Hemley, M. Jackson, F. Spera, S. Redfern, and M. Miao: *Pressure-induced redox reversal of iron and the distribution of elements in deep Earth*. Talk at the 2026 American Physical Society Global Summit (APS), Denver, Colorado, March 2026.
- A. Pandit (presenter), A. Ellis, S. Scott, and M. Miao: *Interactions that Govern the Structure of α -Manganese*. Talk at the 2025 American Physical Society Global Summit (APS), Anaheim, California, March 2025.

HONORS AND AWARDS

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| • College of Science and Mathematics Dean's Scholarship - \$3,000
<i>Northridge, CA</i> | 2026 |
| • APS Global Physics Summit DCOMP Travel Award - \$800
<i>Denver, CO</i> | 2026 |
| • Associated Students Travel Award - \$650
<i>Northridge, CA</i> | 2026 |
| • Second Place, Sigma Xi Research Symposium
<i>Northridge, CA</i> | 2025 |
| • AI4Mat-NeurIPS 2025 Travel Grant - \$500
<i>San Diego, CA</i> | 2025 |

TEACHING EXPERIENCE

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| • Teaching Assistant – Physical Chemistry II (CHEM 352L), CSUN | 2024 – 2025 |
| ◦ Lectured a class of 30 undergraduate for 3 semesters on performing Gaussian calculations in an HPC environment. | |
| ◦ Held walk-in hours for students to receive help with calculations and Linux-environment navigation. | |
| • Teaching Assistant – Computational Molecular and Materials Chemistry (CHEM 555L), CSUN | 2023 |
| ◦ Assisted undergraduate and graduate students with completion of final projects, including calculations on borohydride cages, liquid- and solid-state IR spectra, Raman spectroscopy, and solvent effects. | |

MENTORING AND OUTREACH

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| • Undergraduate Researchers Mentored – Miao Research Group | |
| ◦ S. Ellis – mentored research on compressed hydrogen using hybrid-functional calculations; currently an NSF Fellow pursuing an M.S. in Chemistry at CSUN. | |
| ◦ E. Gonzalez – mentored research on monolayer-to-bulk transitions in transition-metal systems; currently an MCS Associate Quality Control at Amgen. | |
| ◦ L. Wyke – mentored development of machine-learned interatomic potentials for high-pressure systems; currently pursuing an M.S. in Chemistry at CSUN. | |
| • Graduate Researchers Mentored – CSUN Chemistry | |
| ◦ C. Blair – trained in Gaussian calculations for environmental chemistry, transition-state prediction, and reaction kinetics; currently a Ph.D. student at UC Irvine. | |
| ◦ S. Mendoza – trained in applying and fine-tuning MACE-OFF for transition-state prediction; currently an R&D Chemist at ViTrax. | |
| ◦ N. Porcellino – assisted with calculations of high-pressure ytterbium hydride phases; currently a Laboratory Technician at CSL Plasma. | |
| • Vice President and Co-founder – CSUN Chemistry & Biochemistry Journal Club | 2025 – 2026 |
| ◦ Co-founded CSUN's first chemistry and biochemistry journal club and led outreach, article selection, presenter scheduling, and meeting logistics. | |
| ◦ Facilitated paper discussions, figure analysis, and student presentations to strengthen scientific literacy and communication skills. | |
| • Selected Outreach and Service | 2024 – 2026 |
| ◦ Invited talks and panels: Presented "Scientific Conferences: What to Expect" in CHEM 470, discussed the 2025 Nobel Prize in Physics at Chemistry Club's "Breaking Science," and answered undergraduate questions during the Chemistry Club Graduate Panel. | |
| ◦ Research and college-access outreach: Presented computational materials research to visiting high school students during CSUNPosium and volunteered with the Office of Undergraduate Research's "Getting into Research" event to connect students with faculty laboratories. Created LaTeX CV/Resume template used by 30+ students | |
| ◦ K-12 science engagement: Presented research on molecular-hydrogen metallization and tunable 2D electrides during the 2024 and 2025 CSUN Matador Days and introduced local middle school students to high-pressure chemistry during CSUN Science Day. | |